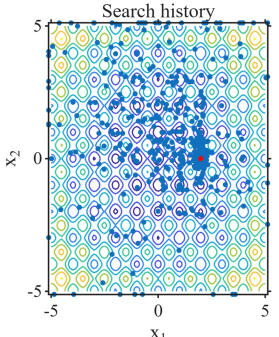
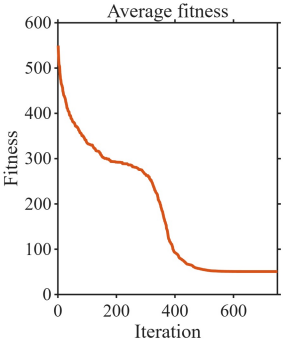
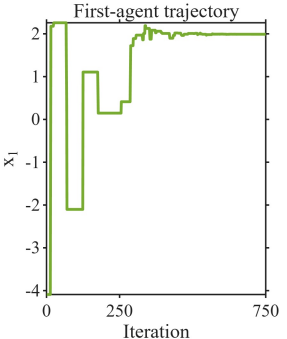
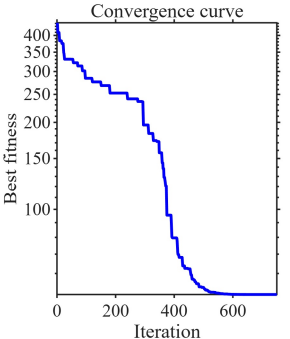

F9
Fitness
80
60
40
20
0
5
0
-5
x_2
-5
0
5
x_1


Search history
x_2
5
0
-5
-5
0
5
x_1


Average fitness
Fitness
600
500
400
300
200
100
0
0
200
400
600
Iteration


First-agent trajectory
x_1
2
1
0
-1
-2
-3
-4
0
250
750
Iteration


Convergence curve
Best fitness
400
350
300
250
200
150
100
0
200
400
600
Iteration